

# Market Making

## Nomura Quant Challenge 5

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# 1 Introduction

A market is fundamentally a mechanism that enables exchange between parties with different needs or preferences. Think of an exchange as an auction, where participants can place bids and offers and auctioneer acts as a matching engine, where "Exchange" occurs when bids and offers match. An asset is any object with a market value. This is where the concept of liquidity comes in. Imagine a hypothetical scenario, where you want to buy an asset (say a pen), but no participant is ready to sell it, we call this "liquidity-crunch" on the sell-side. Liquidity is a measure that shows how easily is the asset tradeable.

Now we define a few terms:

**Definition 1.1.** (*Market Maker*) *Market Maker (also what we will refer as Liquidity Provider/LP later on) is a unit that streams prices/quotes, defining the prices at which they are ready to buy/sell the asset.*

By definition Market-Makers behave as the "directional - agnostic" participants of the market. By directional-agnostic we mean, they are ready to place both bids and offers.

**Definition 1.2.** (*Quotes*) *Quote is basically a triplet  $(V, P, \mathbb{I})$  that says you will buy/sell  $V$  units of an asset at a quote price  $P$  if  $\mathbb{I} = 1/-1$ .*

**Definition 1.3.** (*Bid and Ask*) *The price at which a LP quotes a buy/sell is called the bid/ask price.*

**Definition 1.4.** (*Liquidity Taker*) *Liquidity Taker (also what we will refer as clients later on) is a unit that seeks to take the liquidity provided by the Market Maker for executing the trade.*

The definitions set two different environments that face each other endlessly in the domain of financial markets. The difference in inherent nature of LP and Client defines the mindset of each to earn profit. We now define the term "quote" more formally,

**Definition 1.5.** (*Order Book*) *A long standing list of a series of buy and sell quotes is called the Order Book. Since the number of participants of the market is not limited, the orderbook is dynamic and can have quotes change dynamically.*

A LP's buy quote is basically the price at which LP will buy (i.e, client sells) and vice-versa. From definition it's clear that an order book will have a series of buy

quotes and sell quotes waiting to get executed.

From the client's perspective, the buy quote with the highest bid is the best bid. Similarly from a client's perspective the sell quote with the lowest ask is the best ask.

Now we define a few more terms:

**Definition 1.6.** *(Mid) Mid ( $m$ ) is defined as the average of the best bid ( $b$ ) and best ask ( $a$ ).*

$$m = \frac{b + a}{2} \quad (1)$$

**Definition 1.7.** *(Spread) The difference between the best bid and best ask at any given time is the spread ( $s$ ).*

$$s = a - b \quad (2)$$

As an LP you want the spread to be positive, i.e, "we buy low and sell high". A last definition before we jump into more interesting stuff,

**Definition 1.8.** *(Trade Price) Trade price ( $T_P$ ) is the price at which the trade was executed. (Note this need not necessarily match the mid at that point of time, rather would match with a quote in the orderbook).*

Now we will move to setup a scenario where you have to act like a LP and design strategy to optimize your profit and loss (PnL) margins.

## 2 Scenario Setup For Task 1 and Task 2

The profit and loss (PnL) of a client - LP pair is a zero sum game, in a very simple scenario. Both the parties want to maximise the profit margins. As a Market Maker your job is to have a net - zero inventory at end of the trading session.

**Definition 2.1.** *Net inventory is, at the end of trading session if you have non zero positions in the assets that you traded in, for that particular session*

Excess assets more than your original balance will harm your PnL if prices go down and people buy at cheaper rates from you and vice versa for having fewer assets than original. This is what we will refer to as your **Inventory**.

Now you are given a data set that shows the trade data for a period of 2 months for 6 clients with you as the LP. The schema is as follows:

- **Date:** Date on which trade happened.

- **time:** Timestamp at which trade happened.
- **Name:** Name of the client who executed the trade.
- **Side:** Side of the trade from LP's perspective.
- **Volume:** Total quantity of assets that got traded.
- **Trade Price:** Price at which trade was executed.
- $M_0$ : Mid price of the asset at the time of trade execution.
- **Spread:** Spread ( $S$ ) of the asset order book at time of execution.
- $M_i$ : 6 columns for  $i \in \{5, 10, 15, 20, 25, 30\}$  showing the mid of the asset  $i$  seconds after the trade.

We assume any trade executed will be closed (i.e, the volume you accumulated in your balance will be cleared) within 30 seconds. The mechanism of how this happens need not be specified. For our purpose any loss we get due to this "closing" is a profit for the client that traded with us, and any Profit we realise is a loss for the client. From now on, unless mentioned otherwise, we will act as LP and every quantity will be from our perspective. We now define a few terms,

**Definition 2.2.** (*Side*) We define side from our perspective as,

$$side := \begin{cases} +1 & \text{if LP(we) buy} \\ -1 & \text{if LP(we) sell} \end{cases} \quad (3)$$

**Definition 2.3.** (*PnL*) Lets say weight ( $w_i$ ) of the original trade volume got closed at time  $t = 5i$  seconds, where  $i \in \{1, 2, 3, 4, 5, 6\}$  and  $\sum_{i=1}^{i=6} w_i = 1$ . The PnL of the trade is defined as follows,

$$PnL := (side) * \sum_{i=1}^{i=6} w_i V(M_t - T_P) \quad (4)$$

where  $V$  is the volume of the trade and  $T_P$  is the price at which trade was executed. Note this PnL is from our (LPs) perspective.

**Corollary 1.** (*PnL at closing time  $t = \tau$* ) So when we say the trade was closed at  $t = \tau$ , the PNL from a LP's perspective is written as,

$$PnL(at\ t = \tau) = (side) * V(M_t - T_P) \quad (5)$$

which basically mean we set  $w_i = 1$  for  $i = t/5$  and 0 otherwise in equation 4.

**Corollary 2.** (Aggregate PnL) A direct corollary of 2.3, is **Aggregate PnL**. We define the same as,

$$\text{Aggregate PnL} := (\text{side}) * \sum_{i=1}^{i=6} \frac{V(M_t - T_P)}{6} \quad (6)$$

and by definition this is a single unique metric per trade.

**Definition 2.4.** (Adverse Trade) A trade is said to be adverse at  $t = \tau$ , if the client can close the position at  $t$  with a loss from LP's perspective. More formally,

$$PNL(\text{at } t = \tau) = (\text{side}) * V(M_t - T_P) < 0$$

## 2.1 Task 1 [5 marks]

1. Simulate the adversity profile of each client for each  $\tau \in \{5, 10, 15, 20, 25, 30\}$ . Adversity of a client at a given horizon  $\tau$  is basically percentage of trades of that client, that were labelled as adverse for the given  $\tau$ .
2. Explain the trend, if any, you see in the profiles of each client.

## 2.2 Submission Format for Task 1

Submit a **.py** file containing a Python function with signature:

```
def adversity_profile(client: str, tau: List[int]) -> List[float]:
    """
    Parameters:
        client: Client identifier (single character or string)
        tau: List of horizons e.g. [5, 10, 15, 20, 25, 30]

    Returns:
        List of floats representing adversity percentage at each horizon
    """
```

Submit a **task1\_results.csv** file structured as follows:

| client   | $\tau = 5$ | $\tau = 10$ | $\tau = 15$ | $\tau = 20$ | $\tau = 25$ | $\tau = 30$ |
|----------|------------|-------------|-------------|-------------|-------------|-------------|
| ClientA  | -          | -           | -           | -           | -           | -           |
| ClientB  | -          | -           | -           | -           | -           | -           |
| $\vdots$ | $\vdots$   | $\vdots$    | $\vdots$    | $\vdots$    | $\vdots$    | $\vdots$    |

where each entry is the adversity percentage (0–100) for the given client at the given horizon. Also submit a brief write-up ( $\leq 1$  page), in **.pdf** format, explaining the trend, if any, you see in profiles of each client.

## 2.3 Task 2: Client Profitability and Spread Recommendation

*[5 marks]*

1. For each client, compute the expected PnL per trade (eq. 5) your (LP's) perspective at each horizon  $\tau \in \{5, 10, 15, 20, 25, 30\}$
2. Classify each client as **net-profitable** or **net-costly** to trade with. Justify your classification with quantitative evidence.
3. For each client, recommend a **minimum half-spread**  $\delta_{client}^*$  (in the same units as the data) such that if you had quoted at  $M_0 \pm \delta_{client}^*$  for all their trades, your expected aggregate PnL per trade (eq. 6) against that client would be non-negative. Explain how this quantity relates to the adversity profile from Task 1.

Note: By expected PnL(or expected Aggregate PnL) we mean the sample average of PnL computed per trade.

## 2.4 Submission Format for Task 2

Submit a **.py** file containing Python functions with the following signatures:

```
def expected_pnl(client: str,
                 tau: List[int]) -> dict:
    """
    Parameters:
        client: Client identifier
        tau: List of horizons e.g. [5, 10, 15, 20, 25, 30]

    Returns:
        Dictionary with keys:
            'per_horizon': List[float]
                        Expected PnL per trade at each tau
                        (Corollary 1, eq. 5)
            'aggregate': float
```

```

Expected Aggregate PnL per trade
(Corollary 2, eq. 6)
"""

def classify_client(client: str) -> str:
    """
    Parameters:
        client: Client identifier

    Returns:
        'profitable' or 'costly'
        Based on the aggregate expected PnL (eq. 6)
    """

def min_half_spread(client: str) -> float:
    """
    Parameters:
        client: Client identifier

    Returns:
        Minimum half-spread (in data units) such that
        expected aggregate PnL (eq. 6) >= 0
    """

```

Submit a **task2\_results.csv** file structured as follows:

| <b>client</b> | $\tau = 5$ | $\tau = 10$ | $\tau = 15$ | $\tau = 20$ | $\tau = 25$ | $\tau = 30$ | <b>agg_pnl</b> | $\delta^*$ |
|---------------|------------|-------------|-------------|-------------|-------------|-------------|----------------|------------|
| ClientA       | -          | -           | -           | -           | -           | -           | -              | -          |
| ClientB       | -          | -           | -           | -           | -           | -           | -              | -          |
| $\vdots$      | $\vdots$   | $\vdots$    | $\vdots$    | $\vdots$    | $\vdots$    | $\vdots$    | $\vdots$       | $\vdots$   |

where each  $\tau$  column contains the expected PnL per trade (from the LP's perspective) assuming the trade is closed entirely at  $t = \tau$  (eq. 5), **agg\_pnl** is the expected aggregate PnL per trade using uniform closing weights (eq. 6), and  $\delta^*$  is the minimum half-spread required to make the aggregate PnL non-negative for that client. Also submit a brief write-up ( $\leq 1$  page), in **.pdf** format, containing:

- Classification of each client with quantitative justification.
- Explanation of how  $\delta_{\text{client}}^*$  relates to the adversity profile from Task 1.

### 3 Scenario Setup For Task 3, Task 4 and Task 5

Now as an LP, you can externalize your trade. What do we mean by externalization? Say if a client sells you 100 units of the asset. If you predict the trade can be adverse, then you have the leverage to sell back the same amount (100 units) of asset at the same trade price to some third party. This strategy of risk management is called externalization, where the market maker (or LP) forcibly closes the position. The third party is usually another LP, and for simplicity we assume no extra cost is involved to externalize a trade.

How good your PnL would be, will be directly determined by how good your externalization strategy is. For the current scenario, let's design a naive, yet powerful, externalization strategy.

#### 3.1 The Strategy

For a model  $M$ , let  $p^{+/-}$  denote the probability that a buy/sell trade from LPs (your perspective) will be an adverse trade. Let  $S$  denote the bid-ask spread at the time of trade. Our strategy will externalize the trade iff,

$$p^{+/-} > \theta \quad (7)$$

for buy and sell order respectively, where  $\theta$  is defined as the cutoff probability. The PnL of all the trades that get externalized will exactly balance the PnL of the same trade, had that been internalized. This is a simplified assumption that externalization will net out the (supposed) loss that could have been earned had the trade been internalized. So the overall PnL of an entire trading day  $D$  will look like,

$$PnL_D = \sum PnL_{OfInternalizedTrades} \quad (8)$$

#### 3.2 Task 3

[15 marks]

Design a model  $M$ , that predicts the probability of the trade being adverse for a given client at a given time horizon  $\tau \in \{5, 10, 15, 20, 25, 30\}$ . More formally,

$$M(client, trade, features, \tau) = P(\text{Trade is Adverse at } t = \tau) \quad (9)$$

We call this to be the adversity profile of the client trade. Split the given data into train, validation and test data sets (suggested ratio: 60/20/20 by date).



### 3.3 Submission Format for Task 3

Submit a **.py** file containing your full model training pipeline. Your file **must** expose the following prediction interface:

```
def predict_adversity(*args, **kwargs) -> float:
    """
    Returns:
        Probability in [0, 1] that the trade is adverse at
        the given horizon.

    Note:
        Arguments are left open. Your implementation may
        accept any combination of client, features, tau,
        etc. as needed by your model design.
    """

def compute_metrics(*args, **kwargs) -> pd.DataFrame:
    """
    Returns:
        DataFrame with columns [accuracy, precision,
        recall, log_loss] and rows [train, validation,
        test]. The overall metrics are the average of the
        individual metrics per horizon.
    """
```

Submit a **task3\_results.csv** file structured as follows:

| split      | accuracy | precision | recall | log_loss |
|------------|----------|-----------|--------|----------|
| train      | -        | -         | -      | -        |
| validation | -        | -         | -      | -        |
| test       | -        | -         | -      | -        |

where each entry is the corresponding metric evaluated on that data split. Also submit a clearly documented description of the feature vector ordering (i.e., which index corresponds to which feature). Additionally, submit a brief write-up ( $\leq 2$  pages), in **.pdf** format, covering:

- Choice of model and rationale.
- Feature selection and justification.

### 3.4 Task 4: Optimal Externalization Threshold [15 marks]

For each horizon  $\tau \in \{5, 10, 15, 20, 25, 30\}$ , assume the trade is close at that  $t = \tau$ , now using your model  $M$  from Task 3 and the externalization strategy defined in Section 3.1:

1. Split your data into train, validation, and test sets (suggested ratio: 60/20/20 by date).
2. For a range of cutoff values  $\theta \in [0, 1]$ , compute the total PnL on the validation set under the strategy: externalize trade if  $p^{+/-} > \theta$ , for the given  $\tau$ .
3. Plot  $PnL_{validation}(\theta)$  and identify the optimal threshold  $\theta^*$  that maximizes validation PnL.
4. Discuss:
  - What happens when  $\theta$  is too low (over-externalization)? What revenue do you lose?
  - What happens when  $\theta$  is too high (under-externalization)? What adverse flow do you absorb?
  - Is the optimal  $\theta^*$  the same across all clients, or should it be client-specific? Justify with evidence.
5. Report the final PnL on the held-out test set using your chosen  $\theta^*$  (or client-specific thresholds), for the given  $\tau$ .

### 3.5 Submission Format for Task 4

Submit a **.py** file that **must** contain the following named functions:

```
def optimal_threshold(...) -> dict:
    """
    Returns:
        Dictionary with keys:
            'theta': float or Dict[str, float]
                    Global threshold, or {client: threshold}
            'validation_pnl': float
                    PnL at optimal threshold on validation set
            'test_pnl': float
```

```

                                PnL at optimal threshold on test set
    """

def plot_pnl_vs_theta(...) -> None:
    """
    Plots PnL_validation(theta) for theta in [0, 1].
    Saves figure to 'pnl_vs_theta.png'.
    """

```

The function arguments should include the information for the time horizon ( $\tau$ ) you are computing the metrics for. Apart from the two mentioned function, you can include other functions and the function signatures are left open, so you can include functions/arguments depending on your model design.

Submit a **task4\_results.csv** file structured as follows:

| client   | $\tau$   | $\theta^*$ | final_pnl |
|----------|----------|------------|-----------|
| ClientA  | 5        | -          | -         |
| ClientA  | 10       | -          | -         |
| $\vdots$ | $\vdots$ | $\vdots$   | $\vdots$  |
| ClientB  | 5        | -          | -         |
| ClientB  | 10       | -          | -         |
| $\vdots$ | $\vdots$ | $\vdots$   | $\vdots$  |

where  $\theta^*$  is the optimal externalization threshold for the given client at the given horizon  $\tau$ .

Additionally, submit a brief write-up ( $\leq 2$  pages), in **.pdf** format, containing:

- The  $\text{PnL}_{\text{validation}}(\theta)$  plot.
- Discussion of over-externalization and under-externalization trade-offs.
- Justification for global vs. client-specific thresholds.
- Final test set PnL.

### 3.6 Task 5: Dynamic Quoting Under Inventory Pressure

*[60 marks]*

You now take full control of your quoting strategy. Instead of quoting at fixed spreads, you must design a **quoting function** that dynamically adjusts your bid and ask spreads based on your current state.

**Definition 3.1.** (*Realized Volatility*) The realized volatility  $\sigma_t$  at time  $t$  is defined as:

$$\sigma_t = \sqrt{\frac{1}{N} \sum_{k=1}^N r_k^2} \quad (10)$$

where  $r_k = \frac{M_0^{(k)} - M_0^{(k-1)}}{M_0^{(k-1)}}$  are mid-price returns over the  $N$  most recent trades preceding time  $t$ . Use  $N = 20$ .

**Definition 3.2.** (*Running Inventory*) The inventory  $I_t$  after  $t$  filled trades is:

$$I_t = \sum_{k=1}^t F_k \cdot side_k \cdot V_k \quad (11)$$

where  $F_k \in \{0, 1\}$  is the fill indicator for trade  $k$ .

### 3.6.1 Quoting Function Specification

Design a quoting function:

$$Q(I, \sigma, \alpha, \eta) \rightarrow (\delta_b, \delta_a) \quad (12)$$

where:

- $I \in \mathbb{R}$ : current inventory (positive = long, negative = short)
- $\sigma \in \mathbb{R}_{>0}$ : current realized volatility
- $\alpha \in [0, 1]$ : adversity score from your model  $M$  (Task 3) for the incoming client
- $\eta \in [0, 1]$ : elapsed fraction of the trading day,  $\eta = \frac{t - t_{open}}{t_{close} - t_{open}}$
- $\delta_b \geq 0$ : bid half-spread (distance below mid)
- $\delta_a \geq 0$ : ask half-spread (distance above mid)

Your quoted prices are:

$$b_t = M_t - \delta_b, \quad a_t = M_t + \delta_a \quad (13)$$

**Constraints:**

$$c_{min} \cdot \sigma_t \leq \delta_b, \delta_a \leq \delta_{max} \quad (14)$$

where  $c_{min} = 0.5$  and  $\delta_{max} = 50$  bps of mid. Quotes violating these bounds are clipped.

### 3.6.2 Fill Probability Model

When a trade arrives, the probability that it fills against your quote (rather than going to a competing LP) is:

$$P_{fill}^{side} = \lambda \cdot \exp\left(-\gamma \cdot \frac{\delta_{side}}{\sigma_t}\right) \quad (15)$$

where  $\lambda \in (0, 1]$  is the baseline fill rate and  $\gamma > 0$  is the fill decay parameter.

**Key trade-off:** Wider spreads earn more per trade but fill less often. The parameters  $(\lambda, \gamma)$  are calibrated to the data but **not revealed** to participants.

### 3.6.3 Trade PnL and Inventory Penalty

When a fill occurs at time  $t$ , the trade PnL follows Definition 2.3 with  $T_P^{(t)} = M_t^{(0)} - side_t \cdot \delta_{side_t}$  (you fill at your quoted level). Use uniform closing weights  $w_i = \frac{1}{6}$ .

At the end of each trading day  $D$ , an inventory penalty is applied:

$$\text{Penalty}_D = \phi \cdot I_{T_D}^2 \cdot \sigma_D \quad (16)$$

where  $I_{T_D}$  is end-of-day inventory,  $\sigma_D$  is average daily volatility, and  $\phi > 0$  is a penalty coefficient (not revealed to participants).

### 3.6.4 Regime Shifts

The test stream contains regime-shift points at which  $(\lambda, \gamma, \phi)$  and the adversity characteristics of clients change discontinuously. You are **not told** when or how many regime shifts occur. Your quoting function must adapt through the observable signals  $(I, \sigma, \alpha, \eta)$  alone.

### 3.6.5 Objective and Evaluation

Maximize total net PnL:

$$\text{Objective} = \sum_{D=1}^{T_{days}} \left[ \sum_{t \in D} F_t \cdot PnL_t - \text{Penalty}_D \right] \quad (17)$$

Participants are ranked by a Sharpe-like score:

$$\text{Score} = \frac{PnL_{total}}{\max(\sigma_{portfolio}, \sigma_{floor})} \quad (18)$$

where  $\sigma_{portfolio} = \text{std}(\text{daily PnL})$  and  $\sigma_{floor}$  is a minimum floor. Tiebreaker: maximum drawdown (lower is better).

### 3.6.6 Submission Format for Task 5

Submit a **.py** file that **must** contain the following named functions:

```
def quote(inventory: float, sigma: float,
          alpha: float, eta: float) -> tuple[float, float]:
    """Returns (delta_bid, delta_ask)"""

def validate_quote(...) -> None:
    """
    Runs validation/backtesting of the quoting strategy.
    Prints or saves summary metrics.
    """
```

Function signatures are left open where indicated — structure arguments as needed by your approach. Additionally, submit a brief write-up ( $\leq 1$  page), in **.pdf** format, explaining:

- Your approach and functional form for the quoting strategy.
- Intuition behind the chosen parameterization.
- Validation methodology and results.

## 4 Marking Scheme

| Task                                                   | Marks      |
|--------------------------------------------------------|------------|
| Task 1: Adversity Profile                              | 5          |
| Task 2: Client Profitability and Spread Recommendation | 5          |
| Task 3: Adversity Prediction Model                     | 15         |
| Task 4: Optimal Externalization Threshold              | 15         |
| Task 5: Dynamic Quoting Under Inventory Pressure       | 60         |
| <b>Total</b>                                           | <b>100</b> |